LAL BAHADUR SHASTRI
Group of Institutions

.NET

Lecture-1

"VB. Net"

Presented By:

Jyoti Singh

B.C.A., M.C.A.

BCA-605P

Advanced Technology (Dot Net) Lab

1. Familiarization with IDE.
2. Programming Console applications using VB.NET covering all the aspects of VB.NET fundamentals
3. Object oriented programming using VB.NET covering objects, Inheritance, Polymorphism, Constructors, Static Classes, and Interfaces.
4. Programme to illustrate Exception Handling concepts.
5. Programme to illustrate use of Collections.
6. Programme to perform File I/O Operations.
7. Programming Windows applications using VB.NET covering all major controls and components, Menus, MDI, Event Handling.
8. Creating windows installer.
9. Programme to interact with Database from a Windows Desktop Application.
10. Programming to build web applications using web controls, maintaining state.

Note: The Instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner.

What is VB.NET?

- The VB.NET stands for **Visual Basic. Network Enabled Technologies.**
- It is a simple, high-level, **object-oriented programming** language developed by Microsoft in 2002.
- Furthermore, it supports the OOPs concept, such as abstraction, encapsulation, inheritance, and polymorphism. Therefore, everything in the VB.NET language is an object, including all primitive data types (Integer, String, char, long, short, Boolean, etc.), user-defined data types, events, and all objects that inherit from its base class.
- It is not a case sensitive language, whereas, C++, Java, and C# are case sensitive language.

VB.NET Features

- object-oriented programming language
- design user interfaces for window, mobile, and web-based applications
- It supports Boolean condition for decision making in programming.
- It also supports the multithreading concept, in which you can do multiple tasks at the same time.

.NET Framework

- It is a virtual machine that provide a common platform to run an application that was built using the different language such as C#, VB.NET, Visual Basic, etc.
- The .NET framework is a object oriented, that similar to the Java language. But it is not a platform independent as the Java. So, its application runs only to the windows platform.

- The Main objective of this framework is to develop an application that can run on the windows platform.

- The current version of the .Net framework is 4.8.

Components of .NET Framework

CLR (Common Language Runtime)

CTS (Common Type System)

BCL (Base Class Library)

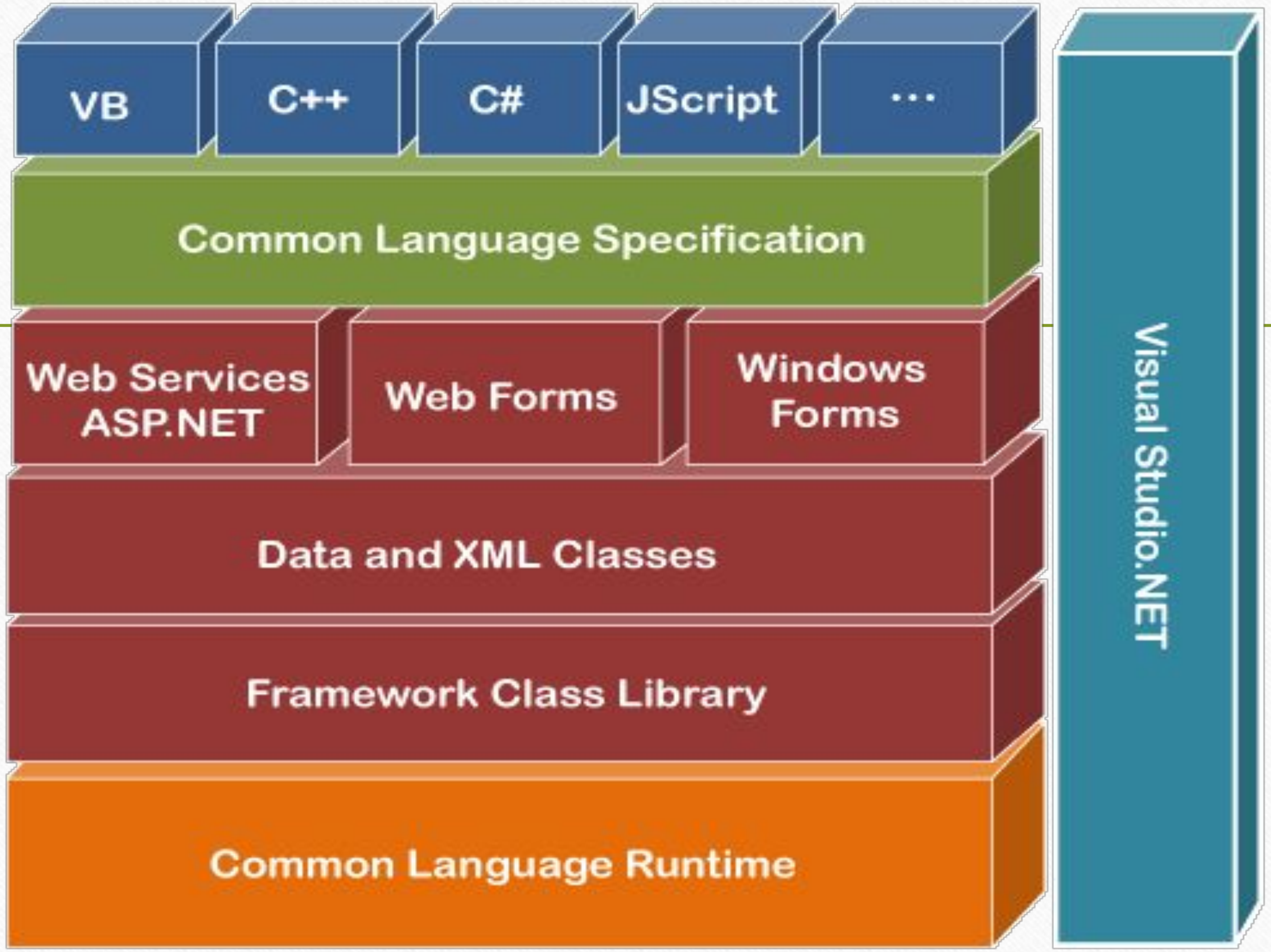
CLS (Common Language Specification)

FCL (Framework Class Library)

.NET Assemblies

XML Web Services

Window Services


VB
C++
C#
JScript
...
Common Language Specification
Web Services
ASP.NET
Web Forms
Windows
Forms
Data and XML Classes
Framework Class Library
Common Language Runtime
Visual Studio.NET

CLR (common language runtime)

- It is an important part of a .NET framework that works like a virtual component of the .NET Framework to executes the different languages program like c#, Visual Basic, etc.
- A CLR also helps to convert a source code into the byte code, and this byte code is known as CIL (Common Intermediate Language) or MSIL (Microsoft Intermediate Language).
- After converting into a byte code, a CLR uses a JIT compiler at run time that helps to convert a CIL or MSIL code into the machine or native code.

CTS (Common Type System)

- It specifies a standard that represent what type of data and value can be defined and managed in computer memory at runtime.
- A CTS ensures that programming data defined in various languages should be interact with each other to share information. For example, in C# we define data type as int, while in VB.NET we define integer as a data type.

BCL (Base Class Library)

- **User defined class library**
- **Assemblies -** It is the collection of small parts of deployment an application's part. It contains either the DLL (Dynamic Link Library) or exe (Executable) file**.**
- **Predefined class library Namespace -** It is the collection of predefined class and method that present in .Net. In other languages such as, C we used header files, in java we used package similarly we used "using system" in .NET, where using is a keyword and system is a namespace.

CLS (Common language Specification)

- It is a subset of common type system (CTS) that defines a set of rules and regulations which should be followed by every language that comes under the .net framework.
- CLS language should be cross-language integration or interoperability.
- For example, in C# and VB.NET language, the C# language terminate each statement with semicolon, whereas in VB.NET it is not end with semicolon, and when these statements execute in .NET Framework, it provides a common platform to interact and share information with each other.

Microsoft .NET Assemblies

- A .NET assembly is the main building block of the .NET Framework.
- It is a small unit of code that contains a logical compiled code in the Common Language infrastructure (CLI), which is used for deployment, security and versioning.
- It defines in two parts (process) DLL and library (exe) assemblies. When the .NET program is compiled, it generates a metadata with Microsoft Intermediate Language, which is stored in a file called Assembly.

FCL (Framework Class Library)

- It provides the various system functionality in the .NET Framework, that includes classes, interfaces and data types, etc. to create multiple functions and different types of application such as desktop, web, mobile application, etc.

Key Components of FCL

Object type

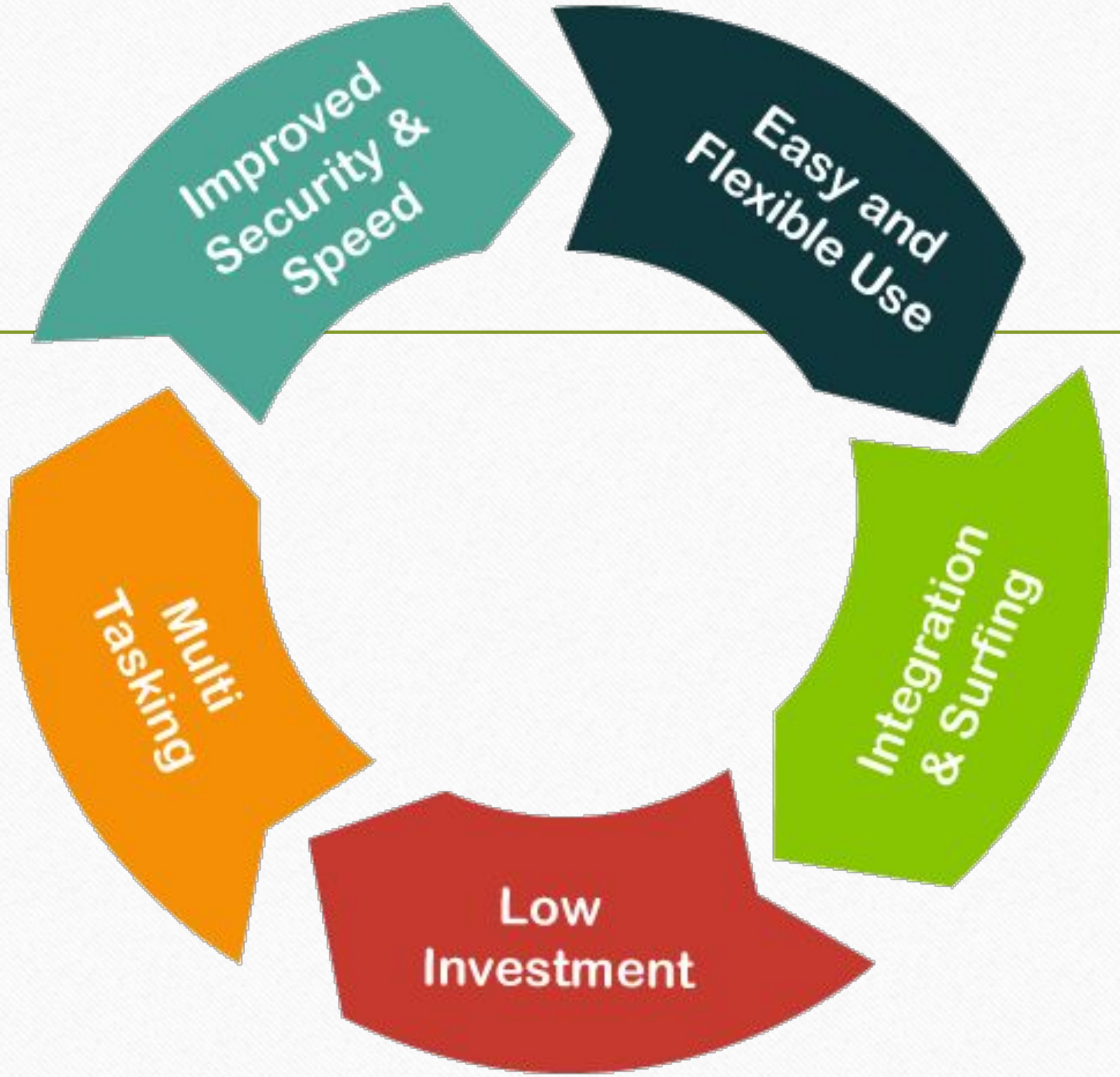
Implementation of data structure

Base data types

Garbage collection

Security and database connectivity

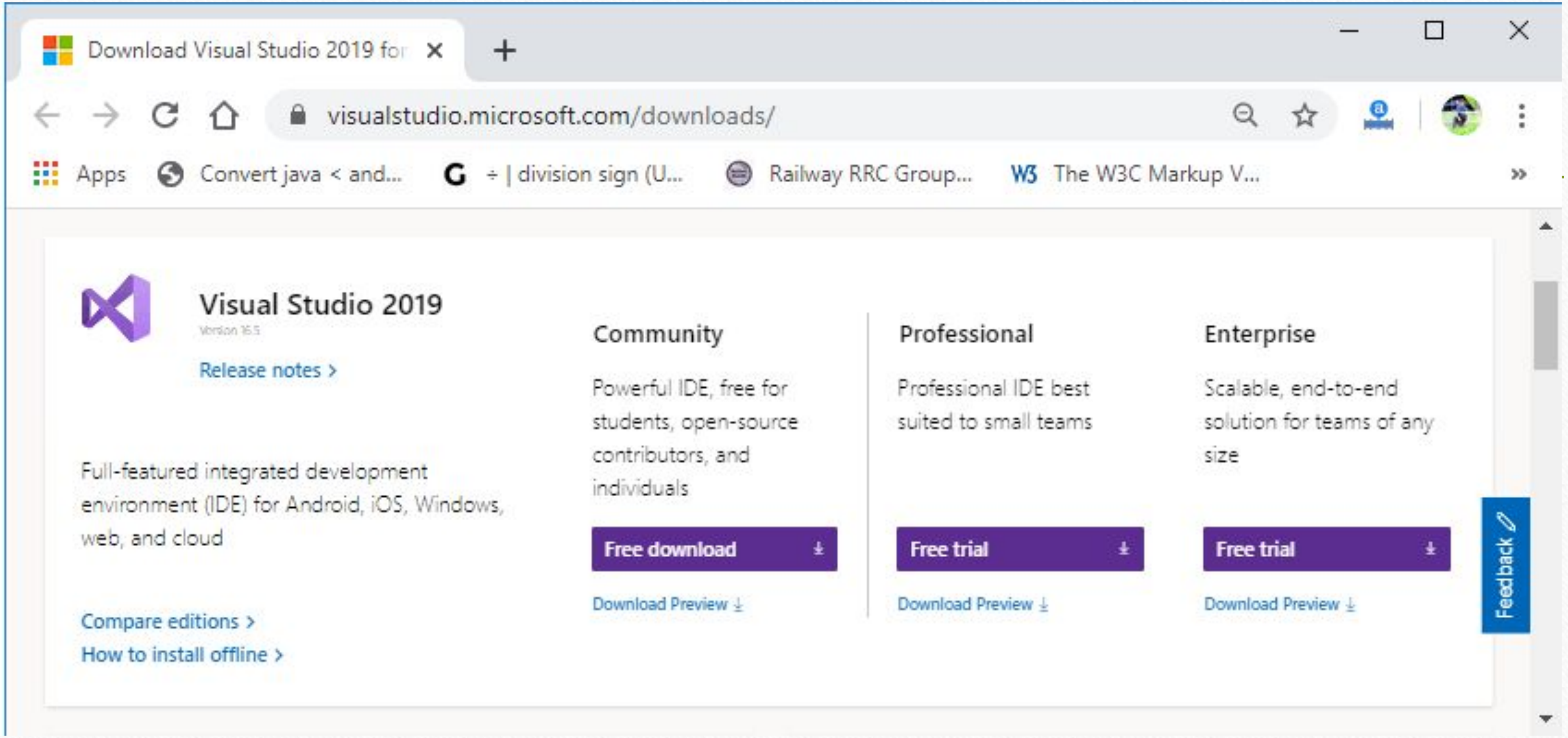
Creating common platform for window and web-based application

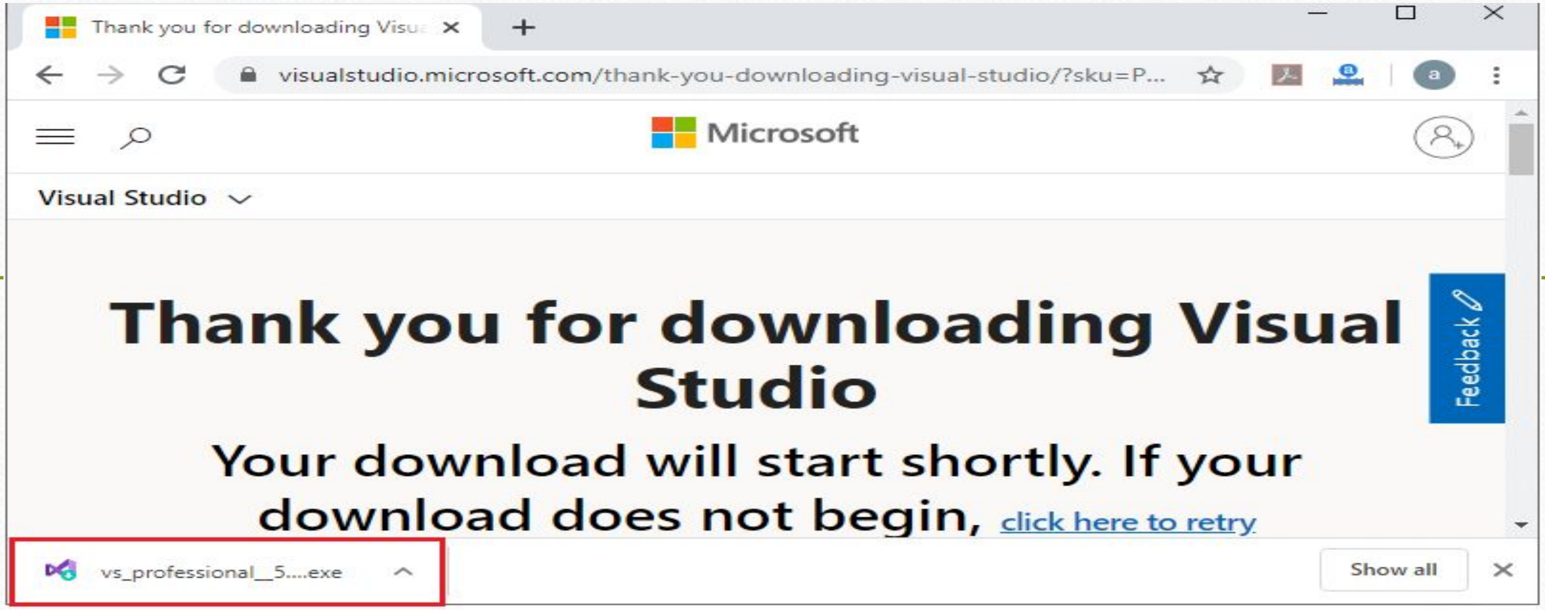

Improved Security & Speed
Easy and Flexible Use
Integration & Surfing
Low Investment
Multi Tasking

- Difference Between VB. NET and Visual Basic
- Difference Between VB.NET and C#
- Difference Between VB.NET and Java

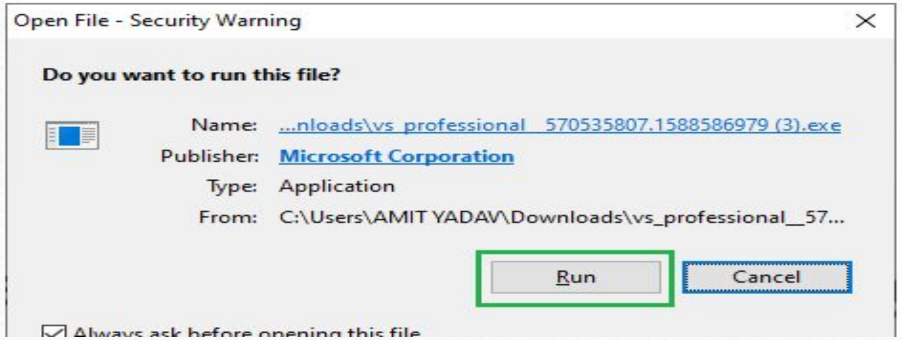
Download and Install Visual Studio

- Visual Studio 2019 was released in 2019 with version number 16.0.
- https://www.visualstudio.com/downloads

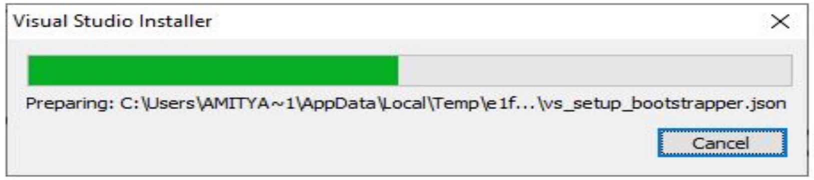

Download Visual Studio 2019 for
visualstudio.microsoft.com/downloads/
Apps
Convert java < and...
÷ | division sign (U...
Railway RRC Group...
The W3C Markup V...
Visual Studio 2019
Release notes >
Full-featured integrated development environment (IDE) for Android, iOS, Windows, web, and cloud
Compare editions >
How to install offline >
Community
Powerful IDE, free for students, open-source contributors, and individuals
Free download
Download Preview
Professional
Professional IDE best suited to small teams
Free trial
Download Preview
Enterprise
Scalable, end-to-end solution for teams of any size
Free trial
Download Preview
Feedback



Step 3. Click on the .exe file and then, it shows a pop-up window.

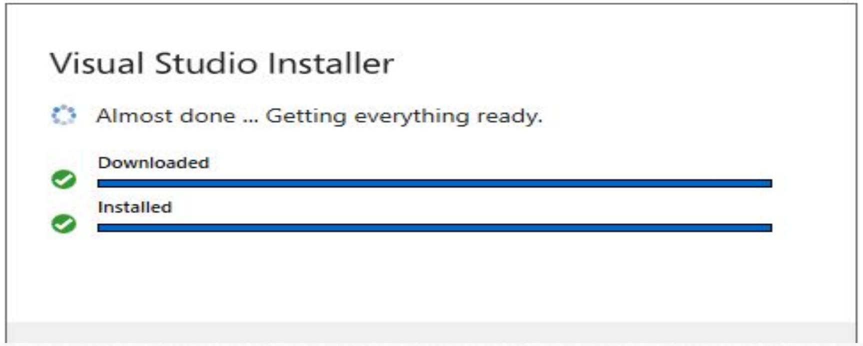


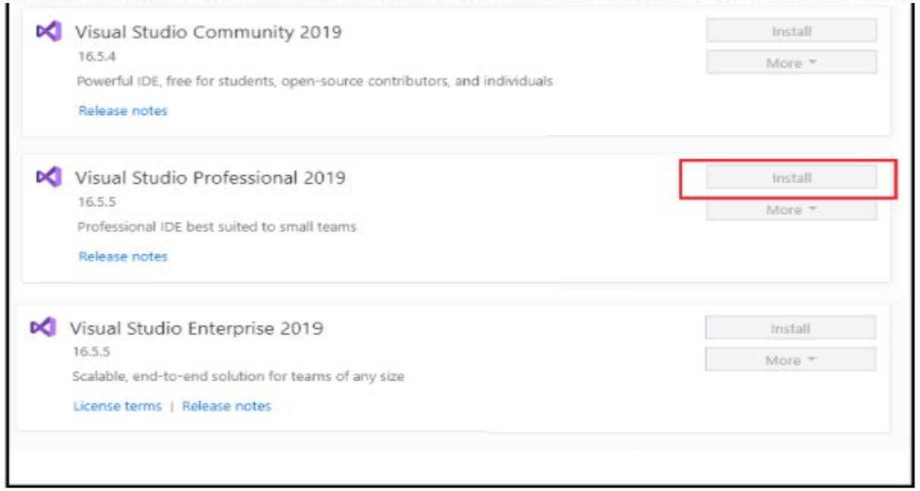
Step 4. Click on the **Run** button, and then it shows the below image.



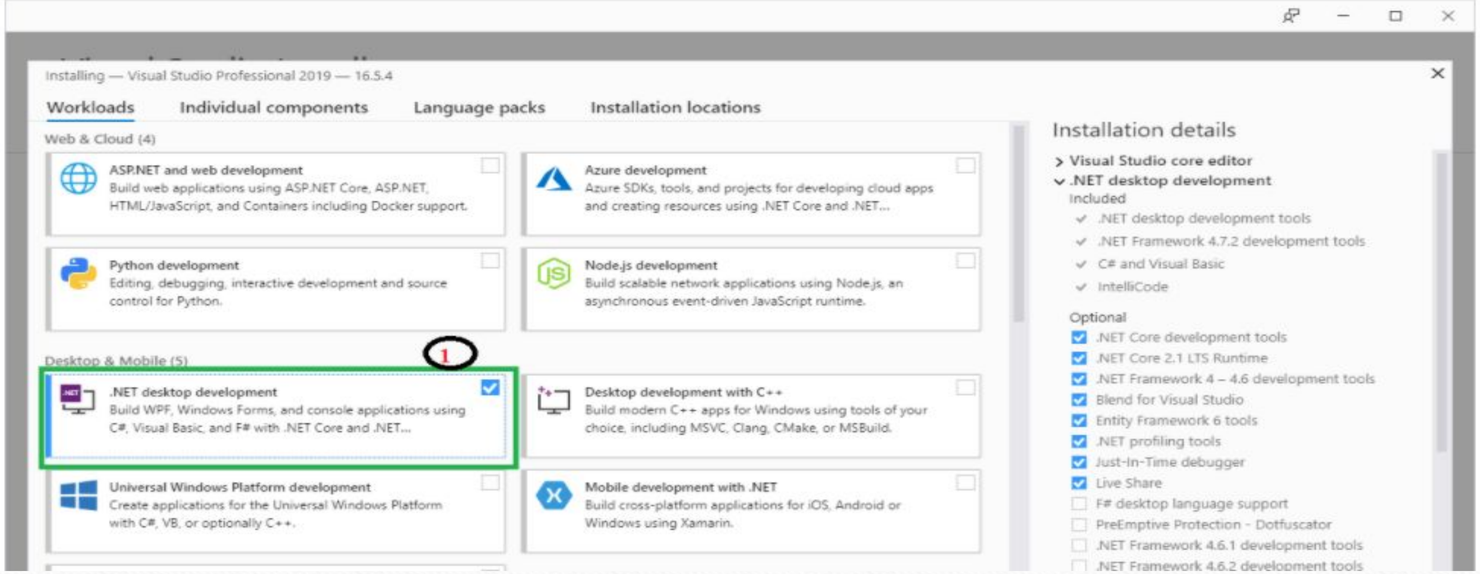
Step 5. Click on the **Continue** Button

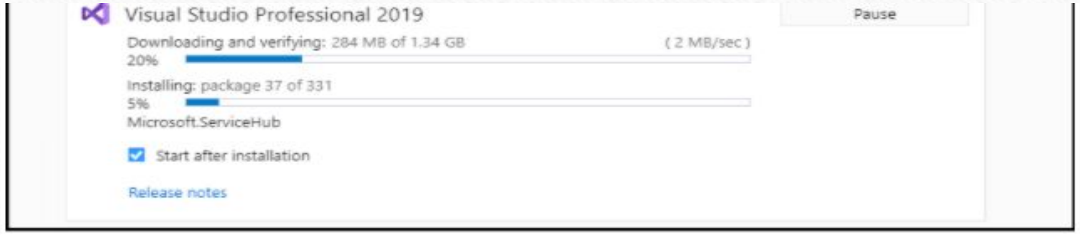
Step 6. After clicking on Continue, Visual Studio will start downloading its initial files as shown in the image below.



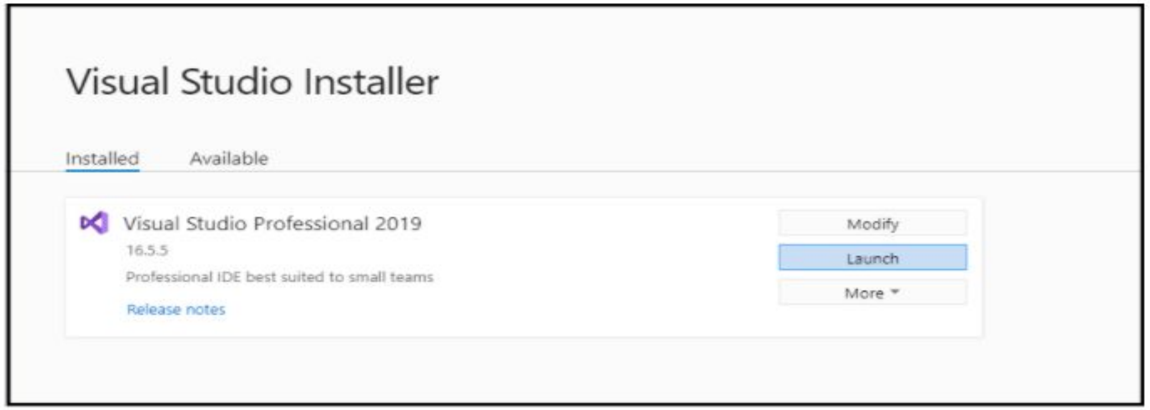


Step 8. After clicking on install button, your Visual Studio IDE will start downloading and then it displays the screen, as we have shown below.

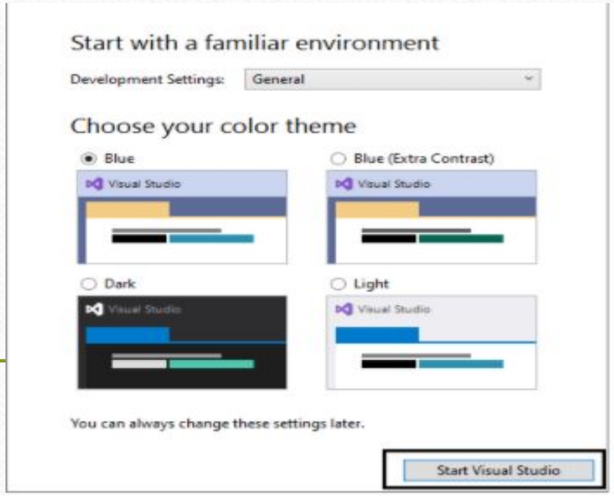




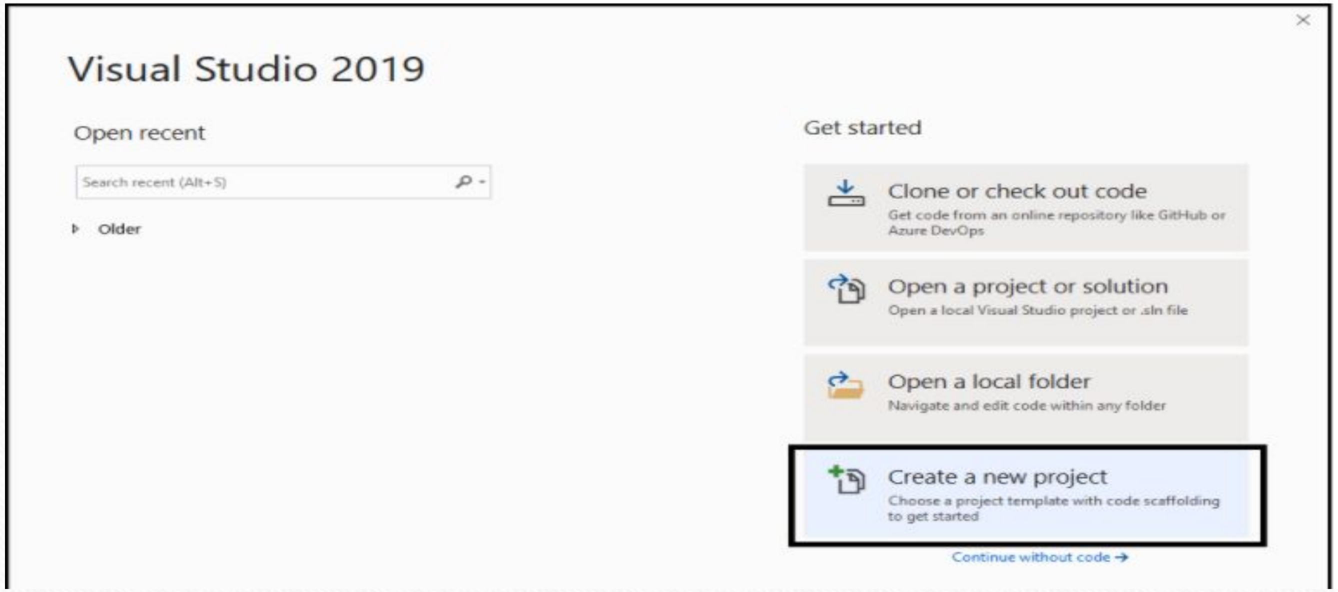
Step 10. After successfully downloading and installing the Visual Studio' supportive file, it shows the below screen in your system.

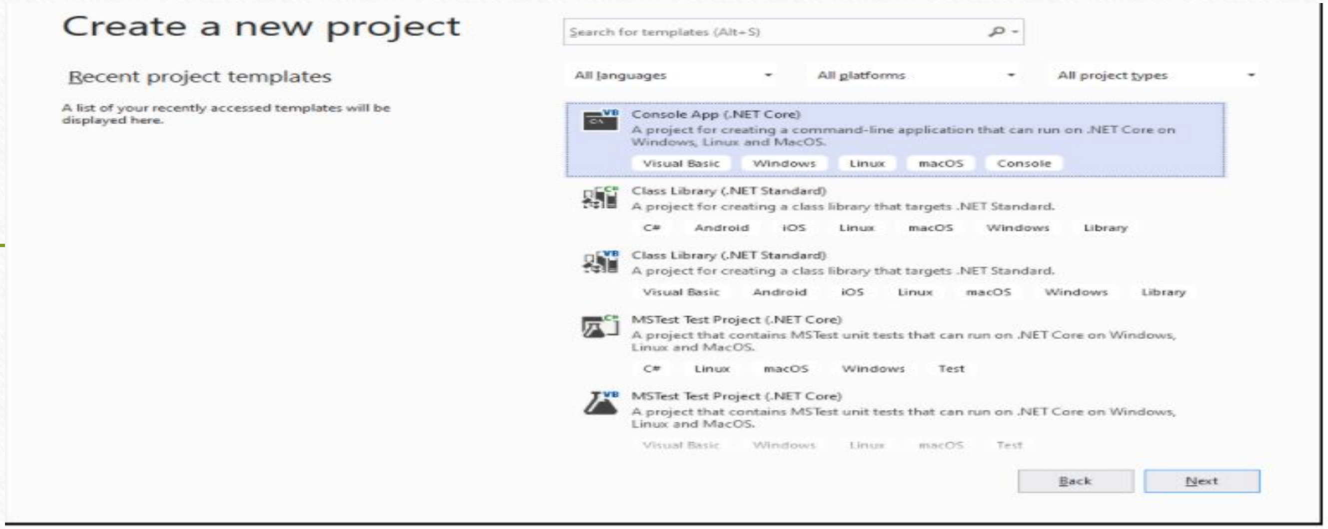


Step 11. Click on Launch button, and then it shows the below image in your screen that represents Visual Studio has been successfully launched in your machine.

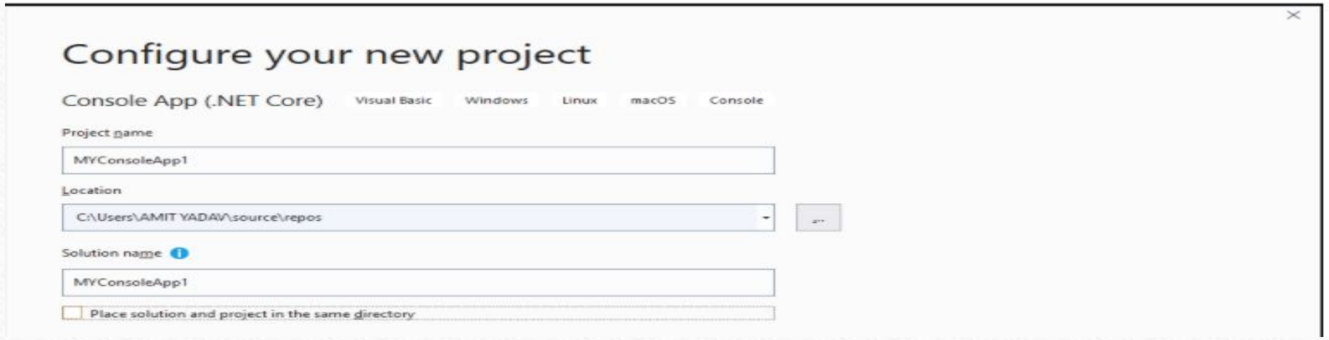


Step 12. After selecting the Theme, click on **Start Visual Studio** and then it shows a below image in your system screen for creating a new project.





Step 14. When you click the **Next** button, it shows the below image to define the project name, and also reminds if you want to place the solution and project in the same directory.



VB.NET Hello World Program

- A VB.NET define the following structure to create a program:
- Namespace declaration
- Procedure can be multiple
- Define a class or module
- Variables
- The Main procedure
- Statement and Expression
- Comments
- Create a **Hello_Program.vb** file in MYConsoleApp1 project and write the following code:

Hello_Program.vb

```
Imports System 'System is a Namespace
Module Hello_Program

    Sub Main()

        Console.WriteLine("Hello, Welcome to the world of VB.NET")
        Console.WriteLine("Press any key to continue...")
        Console.ReadKey()

    End Sub

End Module
```

Let's compile and run the above program by pressing the F5 key, we get the follwoiing output.

Output:

- In VB.NET programming, the first line of the program is "Import System", where Imports is a statement that inherit the system namespace. A System is a namespace that contains basic classes, reference data types, events, attributes, and various inbuilt functions that help to run the program.

- The Second line defines the Module It specifies the VB.NET filename. As we know, VB.NET language is a completely object-oriented language, so every program must contain a module or class in which you can write your program that contains data and procedures within the module.

```
Module Module1
End Module
```

- You can define more than one procedure in classes and modules. Generally, the procedure contains executable code to run. A procedure may contain the following function:

- Function
- Operator
- Sub
- Get
- Set
- AddHandler
- RemoveHandler

- Every program must contain a Main() method. In VB.NET, there is a Main() method or procedure that represents the starting point to execute a program, as we have seen in C language, their entry point is the main() function.
- A comment (') symbol is used to comment on a line that is ignored by the compiler in a program, and it is a good practice to use comments for a better understanding of the program.
- The Console.WriteLine() is a method of the console class. It is used to print any text or messages in the application. And the Console.ReadKey() is used to read a single character from the keyboard back to the Visual Studio IDE.

Thank You